

Problems Encountered During the Project

1. Splitting a Single Dataset into Two Different Source Types

- The original Online Retail dataset was available as a single CSV file. To satisfy the project requirement of integrating two different source types, the dataset had to be separated into a Delta table for product information and JSON files for order data while maintaining a common join key.

2. Configuring Auto Loader for Incremental Processing

- Setting up Auto Loader required proper configuration of schema and checkpoint locations. Ensuring that only newly added files were processed during subsequent runs required careful testing and validation.

3. Joining Data from Different Sources

- The project involved combining data from a Delta table and a streaming JSON source. Ensuring successful joins using the stockcode field and maintaining data consistency across both sources required additional verification.

4. Validating Incremental Updates

- Demonstrating that newly added records correctly propagated through the Bronze, Silver, and Gold layers and appeared in the dashboard required multiple test executions and result verification.